

Comparing Health Events in Populations: A Framework for Analysis

Eric Delmelle



Introduction

- **Definition of Health Events:** Disease outbreaks, chronic conditions, injuries, and health behaviors.
- **Importance of Comparisons:** Understanding disparities, identifying risk factors, guiding public health interventions.
- **Key Concepts:** Population health, epidemiology, and biostatistics.

Objectives of Population Health

Four Key Objectives:

1. **Describe:** Understand population-level health outcomes.

2. **Explain:** Identify determinants and drivers of health outcomes.
 3. **Predict:** Anticipate future health trends and patterns.
 4. **Control:** Implement interventions to improve outcomes.
-

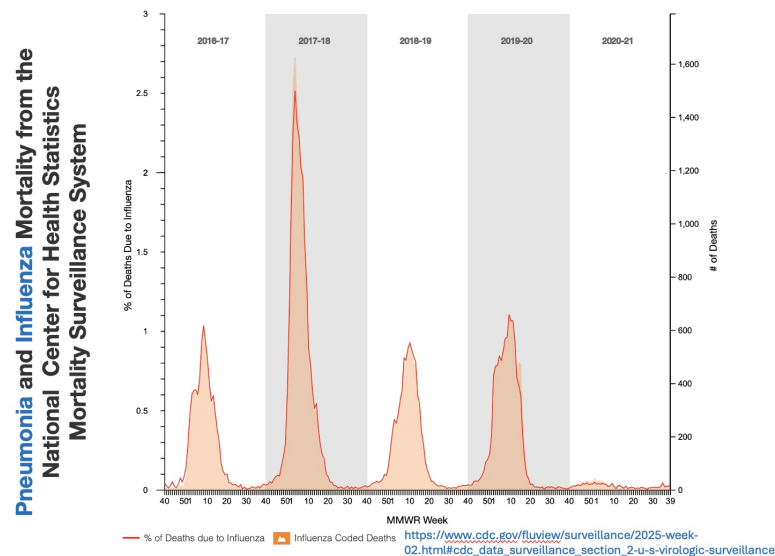
Historical Context

Key Figures:

- **John Snow:** Cholera outbreak mapping.
 - **Ignaz Semmelweis:** Importance of handwashing.
 - **Joseph Goldberger:** Nutritional causes of pellagra.
-

Type of Comparisons

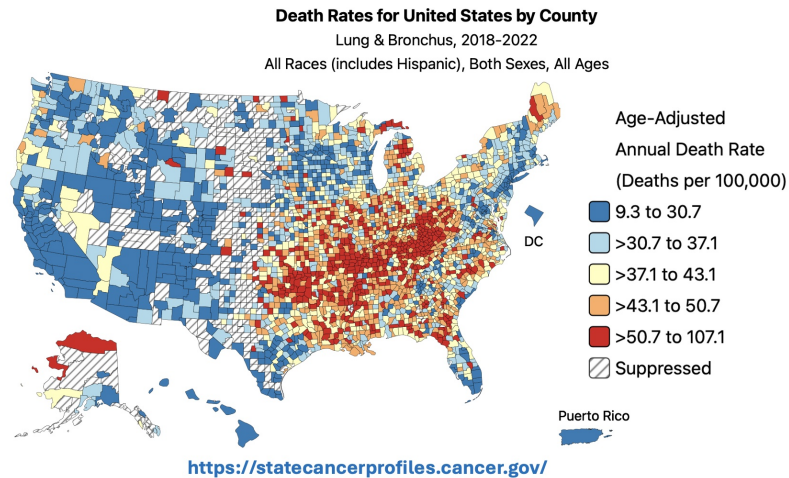
Time-Based



Key Metrics:

- **Incidence:** New cases over time.
 - **Prevalence:** Existing cases at a given time.
-

Place-Based



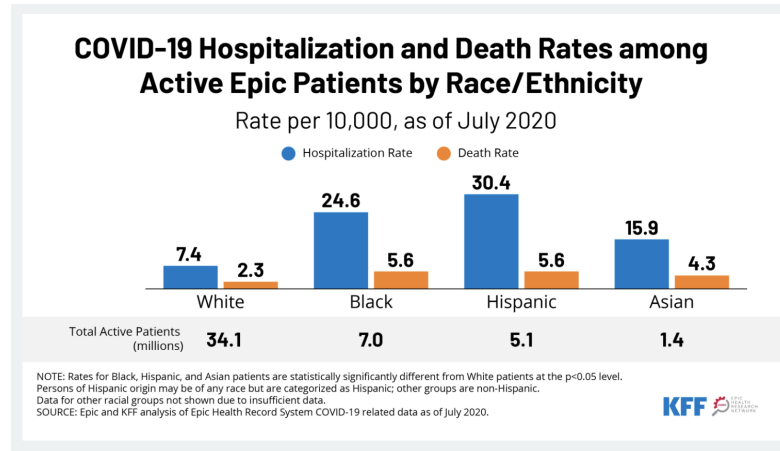
Example:

- **Urban vs. Rural Heart Disease Mortality:**
 - Urban: 50 per 100,000.
 - Rural: 75 per 100,000.
-

Group-Based

Example:

- Health disparities by race, age, and income.



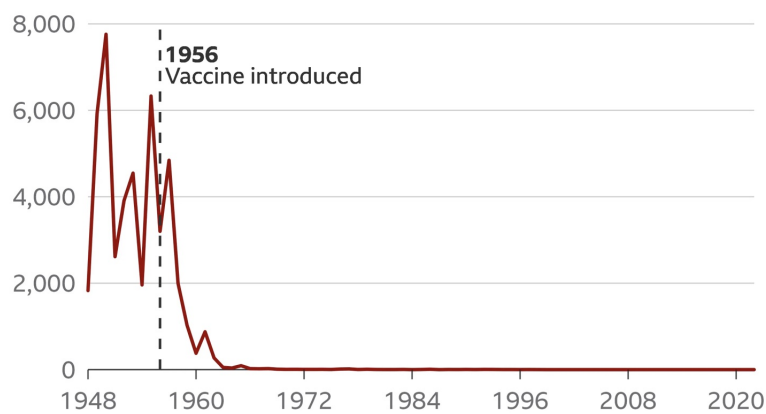
Event-Based

Key Concept:

- Natural experiments: Before vs. after policy changes or interventions.

How polio was eradicated

Number of polio cases in England and Wales (1948-2022)



Source: Public Health England

B B C

Additional Event-Based Example



Measures of Comparison

Key Metrics:

- **Age-Standardized Rates:** Adjusted to eliminate age structure differences.
- **Attributable Risk:** Measures the impact of specific risk factors on outcomes.

Levels of Analysis

Frameworks:

- **Individual-Level:** Biostatistical and clinical trials.
- **Population-Level:** Geographic and demographic patterns.

Determinants of Health

Categories:

1. **Social and Economic Factors**
 2. **Environmental Conditions**
 3. **Behavioral and Genetic Influences**
-

Challenges in Comparisons

Key Challenges:

- **Data Quality:** Inaccuracies or incomplete datasets.
 - **Ethical Considerations:** Privacy and fair comparisons.
-

Population vs. Community Health Assessments

Key Differences:

- **Community Health Assessments:**
 - Focus on local needs/resources.
 - Qualitative methods (e.g., interviews).
 - **Population Health Assessments:**
 - Broad, systemic focus.
 - Quantitative data (e.g., chronic disease rates).
-

Population vs. Community Health Assessments

Example:

- Community: Identifying food deserts.
 - Population: Obesity prevalence across counties.
-

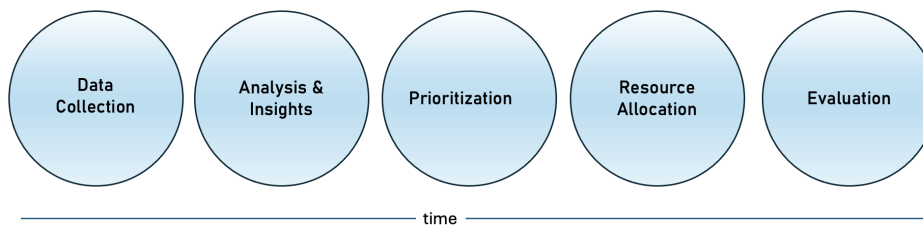
Policy Implications

Using Comparisons to Drive Change:

1. **Set Priorities:** Identify at-risk groups (e.g., elderly, low-income communities).
 2. **Develop Interventions:** Targeted programs (e.g., tobacco cessation).
 3. **Advocate for Policy Change:** Use data for systemic reforms.
-

Policy Implications

Visual:



Interactive Example

Dataset Example:

Population	Cases	Rate (per 100,000)
Urban	200	50
Rural	300	75

Prompt:

- “What does this suggest about resource allocation?”
-

Recap and Transition

Key Takeaways:

- Importance of describing, explaining, predicting, and controlling health events.
- Tools and methods to compare health outcomes.
- Practical implications for population health strategies.

Next:

- Group activity: Apply concepts to a real-world health disparity.
-

Group Activity: Population Health Comparison

Objective:

Apply Chapter 1 metrics to analyze health disparities.

Instructions:

1. Form groups of 5 (see groupe in 2 slides).
-

Instructions:

2. Analyze the provided dataset on coursesite
 - Calculate rates (e.g., incidence, prevalence).
 - Identify disparities (e.g., geographic, demographic).
 - Propose targeted interventions.
 3. Prepare to present findings in 3 minutes.
-

Group Assignments and Analysis Instructions**Analysis Instructions**

- **Analyze the provided dataset on coursesite:**
 - Calculate rates (e.g., incidence, prevalence).
 - Identify disparities (e.g., geographic, demographic).
 - Propose targeted interventions.
-